

## Solution Requirements

|                      |                                                 |
|----------------------|-------------------------------------------------|
| <b>Date</b>          | 11 June 2026                                    |
| <b>Team ID</b>       |                                                 |
| <b>Project Name</b>  | EcoTrack - Personal Carbon AI Footprint Tracker |
| <b>Maximum Marks</b> | 4 Marks                                         |

### Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and

#### 1. Functional Requirements Matrix:

- FR-1: Real-Time Emissions Calculator recording transport distance (km), diet type, food waste, and electricity (kWh).
- FR-2: Groq LLaMA 3.3 AI Recommendation Engine providing personalized lifestyle eco-swaps with CO2 savings.
- FR-3: 30-Day Historical Analytics Dashboard rendering Chart.js line trends, category breakdown charts, and heatmaps.
- FR-4: Gamification System computing logging streaks and unlocking 10 distinct achievement badges.
- FR-5: Firebase Firestore Data Synchronization & Firebase Auth User Authentication.

#### 2. Non-Functional Requirements Matrix:

- NFR-1 (Performance): Aggregate backend API endpoints respond in < 180ms.
- NFR-2 (Availability): Application architecture ensures 99.9% uptime and availability.
- NFR-3 (Usability): ~~Responsive glassmorphism UI layout supporting desktop, tablet, and mobile viewports~~
- NFR-4 (Security): API keys (Groq, Carbon Interface, Firebase) secured via server-side .env environment variables.

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|----------|-----------------------------------|---------------------------------------------------------------------------------------------------------------|--|
|          |                                   | <i>services the system must connect to]</i>                                                                   |  |
| <b>4</b> | Transactions processing           | <i>[Explain how data exchanges, payments, or core system transactions will be handled]</i>                    |  |
| <b>5</b> | Reporting                         | <i>[Describe the types of reports, dashboards, or data exports the system must generate]</i>                  |  |
| <b>6</b> | Business rules, etc.              | <i>[List specific logic, constraints, or workflows unique to the business process]</i>                        |  |
| <b>7</b> | Compliance to laws or regulations | <i>[Specify any legal, industry, or data privacy standards the system must adhere to (e.g., GDPR, HIPAA)]</i> |  |
| <b>8</b> | <i>Other</i>                      | <i>[Add any additional functional requirements here]</i>                                                      |  |

**Step 2: Non-Functional Requirements (NFR)**

| <b>S.No</b> | <b>NFR Category</b> | <b>Requirement Description</b> | <b>Target Metric / Acceptance Criteria</b> |
|-------------|---------------------|--------------------------------|--------------------------------------------|
|             |                     |                                |                                            |

|          |                            |                                                                                              |                                                       |
|----------|----------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>1</b> | Performance & Speed        | <i>[Describe response times, processing speed, etc.]</i>                                     | <i>(e.g., Page loads in &lt; 2 seconds)</i>           |
| <b>2</b> | Scalability                | <i>[Describe how the system handles growth in users or data]</i>                             | <i>(e.g., Supports up to 10,000 concurrent users)</i> |
| <b>3</b> | Security & Data Privacy    | <i>[Detail data protection, encryption, and vulnerability management]</i>                    | <i>(e.g., End-to-end AES-256 encryption)</i>          |
| <b>4</b> | Reliability & Availability | <i>[Specify uptime requirements and fault tolerance]</i>                                     | <i>(e.g., 99.9% uptime per month)</i>                 |
| <b>5</b> | Usability & Accessibility  | <i>[Describe user experience standards and accessibility compliance]</i>                     | <i>(e.g., Complies with WCAG 2.1 AA standards)</i>    |
| <b>6</b> | Other                      | <i>[Add any additional non-functional requirements (e.g., Maintainability, Portability)]</i> |                                                       |